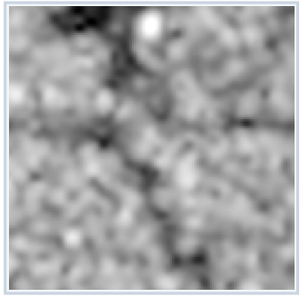
Visual Demonstration

End-to-End Multi-Modal AI Pipeline Execution on Validation Scene: Hurricane Florence (Tile ID: 00000050)

1 Stage 1 • SAR Radar Ingestion

SENTINEL-1 C-BAND



AI Annotation: Radiometrically calibrated C-band SAR backscatter (mean: -9.66 dB).

Field Meaning: All-weather sensing penetrates rain & storm clouds where optical sensors fail.

2 Stage 2 • Flood Inundation Mask

FLOOD U-NET (SAR)

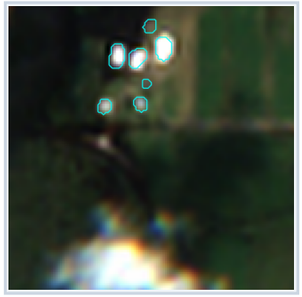


AI Annotation: Segmented 407 flooded pixels (2.48% tile area) from specular radar reflection.

Field Meaning: Maps exact open floodwater boundaries and active water intrusion corridors.

3 Stage 3 • Building Localization

xBD U-NET (OPTICAL)

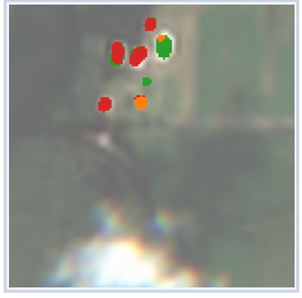


AI Annotation: Segmented 7 residential structures with polygonal footprint boundaries.

Field Meaning: Isolates inhabited structures to establish baseline targets for tactical rescue.

4 Stage 4 • Damage Classification

xBD DAMAGE CLASSIFIER

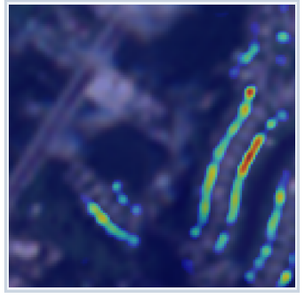


AI Annotation: Classified structural state: 1 Intact (green), 3 Minor (orange), 3 Destroyed (red).

Field Meaning: Identifies roof/wall collapse independent of surrounding standing flood depth.

5 Stage 5 • Explainable AI (XAI)

TORCH-CAM (GRAD-CAM)

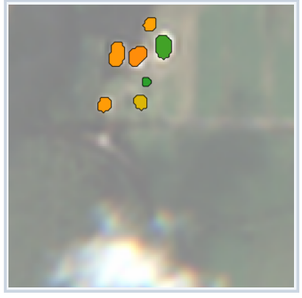


AI Annotation: Visualizes convolutional activations focusing on debris piles & fractured edges.

Field Meaning: Provides explainable audits for commanders before deploying tactical squads.

6 Stage 6 • Multi-Modal Risk Fusion

FUSION CORE (Ri = 0.5Ei+0.5Si)



AI Annotation: Fused flood exposure (Ei) + damage severity (Si) into score Ri (Peak: 0.589).

Field Meaning: Generates priority triage dispatch: 3 High Risk (Ri > 0.50), 2 Moderate, 2 Low.